Date & Time Functions

Excel vs Microsoft SQL Server vs Power BI (DAX) | Study Guide

Purpose: A side-by-side reference for learning date/time functions used in spreadsheet analysis, SQL queries, and Power BI reporting. This guide excludes the practical sales-measures section.

1. Common date and time functions compared

Purpose	Excel	SQL Server	Power BI (DAX)
Create a date	DATE()	DATEFROMPARTS()	DATE()
Current date	TODAY()	CAST(GETDATE() AS date)	TODAY()
Current date and time	NOW()	GETDATE()	NOW()
Extract year / month / day	YEAR() / MONTH() / DAY()	YEAR() / MONTH() / DAY()	YEAR() / MONTH() / DAY()
Extract hour / minute / second	HOUR() / MINUTE() / SECOND()	DATEPART(hour/minute/second, ...)	HOUR() / MINUTE() / SECOND()
Weekday number	WEEKDAY()	DATEPART(weekday, ...)	WEEKDAY()
Week number	WEEKNUM()	DATEPART(week, ...)	WEEKNUM()
Quarter number	ROUNDUP(MONTH(A2)/3,0)	DATEPART(quarter, ...)	QUARTER()
Add days / months / years	A2+days; EDATE()	DATEADD()	Date + days; EDATE()
End of month	EOMONTH()	EOMONTH()	EOMONTH()
Difference between dates	DAYS() / DATEDIF()	DATEDIFF()	DATEDIFF()
Convert text to date/time	DATEVALUE() / TIMEVALUE()	CAST() / CONVERT()	DATEVALUE() / TIMEVALUE()
Create a time	TIME()	TIMEFROMPARTS()	TIME()

Remember: Similar names do not always mean identical behavior. Check argument order, return type, week-number settings, and SQL Server version before replacing one function with another.

2. Excel date and time functions

Principal built-in worksheet functions in Excel's date/time category:

Function	Purpose
DATE()	Create a date from year, month, and day.
DATEDIF()	Calculate difference in years, months, or days.
DATEVALUE()	Convert date text to a date serial value.
DAY()	Extract the day of the month.
DAYS()	Calculate days between two dates.
DAYS360()	Calculate days using a 360-day year convention.
EDATE()	Add or subtract months.
EOMONTH()	Return the last day of a month.
HOUR(), MINUTE(), SECOND()	Extract time components.
ISOWEEKNUM()	Return ISO week number.
MONTH(), YEAR()	Extract month or year.
NETWORKDAYS()	Count working days excluding weekends and optional holidays.
NETWORKDAYS.INTL()	Count working days using a custom weekend pattern.

Function	Purpose
NOW(), TODAY()	Return current date/time or current date.
TIME(), TIMEVALUE()	Create a time or convert time text to a time value.
WEEKDAY(), WEEKNUM()	Return weekday number or week number.
WORKDAY()	Find a date after/before a number of working days.
WORKDAY.INTL()	Workday calculation with customizable weekends.
YEARFRAC()	Calculate the fraction of a year between two dates.

Excel examples

Assume A2 contains 15-Oct-2026 and B2 contains another date:

Requirement	Formula
Extract year	=YEAR(A2)
Extract month	=MONTH(A2)
Extract day	=DAY(A2)
Last day of the month	=EOMONTH(A2,0)
Date three months later	=EDATE(A2,3)
Working days between dates	=NETWORKDAYS(A2,B2)
Current date	=TODAY()
Current date and time	=NOW()

3. Microsoft SQL Server date and time functions

SQL Server functions help extract date parts, calculate intervals, manipulate dates, validate values, and work with time-zone offsets.

Function	Purpose
DATEFROMPARTS()	Create a date from year, month, and day.
DATETIMEFROMPARTS()	Create a datetime from date and time parts.
TIMEFROMPARTS()	Create a time from individual parts.
GETDATE()	Current server date and time (datetime).
CURRENT_TIMESTAMP	Current server date and time; equivalent in purpose to GETDATE().
SYSDATETIME()	Current date/time with higher fractional-second precision.
GETUTCDATE(), SYSUTCDATETIME()	Current UTC date/time; the latter has higher precision.
SYSDATETIMEOFFSET()	Current date/time with time-zone offset.
CURRENT_DATE	Current date only in SQL Server versions that support it.
YEAR(), MONTH(), DAY()	Extract year, month, or day.
DATEPART()	Return a numeric date/time part.
DATENAME()	Return a text name for a date part, such as month name.
DATEADD()	Add or subtract a date/time interval.
DATEDIFF()	Count date-part boundaries between two dates.
DATEDIFF_BIG()	Like DATEDIFF with a larger return range.
EOMONTH()	Return the last day of a month.
DATETRUNC()	Truncate date/time to a boundary (version-dependent).
DATE_BUCKET()	Group dates into fixed-size intervals (version-dependent).
ISDATE()	Test whether an expression is a valid date/time.
SWITCHOFFSET()	Change the offset of a datetimeoffset value.
TODATETIMEOFFSET()	Associate an offset with a date/time value.

SQL Server example

```
SELECT
GETDATE() AS CurrentDateTime,
YEAR(OrderDate) AS OrderYear,
MONTH(OrderDate) AS OrderMonth,
DAY(OrderDate) AS OrderDay,
DATEPART(QUARTER, OrderDate) AS OrderQuarter,
DATENAME(MONTH, OrderDate) AS MonthName,
EOMONTH(OrderDate) AS MonthEnd,
DATEADD(MONTH, 1, OrderDate) AS NextMonth
FROM FactSales;
```

Version note: Function availability varies by SQL Server version. Verify support for DATETRUNC(), DATE_BUCKET(), and CURRENT_DATE in your environment.

4. Power BI date and time functions (DAX)

A. Date and time functions

Function	Purpose
CALENDAR()	Create a continuous date table between two dates.
CALENDARAUTO()	Generate a date table based on date values in the model.
DATE(), DATEVALUE()	Create a date or convert text to a date.
DATEDIFF()	Calculate date-part boundaries between two dates.
DAY(), MONTH(), YEAR()	Extract date components.
EDATE(), EOMONTH()	Shift months or return month-end.
HOUR(), MINUTE(), SECOND()	Extract time components.
NETWORKDAYS()	Count working days.
NOW(), TODAY()	Return current date/time or current date.
QUARTER()	Extract the quarter number.
TIME(), TIMEVALUE()	Create a time or convert time text.
UTCNOW(), UTCTODAY()	Return current UTC date/time or date.
WEEKDAY(), WEEKNUM()	Return weekday or week number.
YEARFRAC()	Calculate the fraction of a year between dates.

B. Time intelligence functions

These functions support period-to-date calculations, comparisons with previous periods, and period-end balances. Some week-based functions require newer calendar-based time-intelligence capabilities; availability depends on model setup and Power BI version.

Function	Purpose
DATEADD()	Shift the selected date period forward or backward.
DATESBETWEEN()	Return dates between specified start and end dates.
DATESINPERIOD()	Return dates in a specified interval.
DATESMTD(), DATESQTD(), DATESYTD()	Return month-, quarter-, or year-to-date date sets.
DATESWTD()	Return week-to-date date set where supported.
FIRSTDATE(), LASTDATE()	Return first or last date in the current context.
PREVIOUSDAY(), PREVIOUSMONTH()	Return the previous day or month.
PREVIOUSQUARTER(), PREVIOUSYEAR()	Return the previous quarter or year.
NEXTDAY(), NEXTMONTH()	Return the next day or month where supported.
NEXTWEEK(), NEXTQUARTER(), NEXTYEAR()	Return next periods where supported.
PARALLELPERIOD()	Return a parallel period shifted by months, quarters, or years.
SAMEPERIODLASTYEAR()	Return corresponding dates in the previous year.
STARTOFWEEK(), STARTOFMONTH()	Return the start of a week or month where supported.
STARTOFQUARTER(), STARTOFYEAR()	Return the start of a quarter or year.
ENDOFWEEK(), ENDOFMONTH()	Return the end of a week or month where supported.
ENDOFQUARTER(), ENDOFYEAR()	Return the end of a quarter or year.
TOTALWTD(), TOTALMTD()	Evaluate an expression week-to-date or month-to-date.
TOTALQTD(), TOTALYTD()	Evaluate an expression quarter-to-date or year-to-date.
CLOSINGBALANCEWEEK(), CLOSINGBALANCEMONTH()	Evaluate an expression at week-end or month-end where supported.

Function	Purpose
CLOSINGBALANCEQUARTER(), CLOSINGBALANCEYEAR()	Evaluate an expression at quarter-end or year-end.
OPENINGBALANCEMONTH(), OPENINGBALANCEQUARTER()	Evaluate an expression at the start of a period.
OPENINGBALANCEYEAR()	Evaluate an expression at the start of the year.

Reference: Microsoft Learn — DAX date and time functions and DAX time intelligence functions. Consult the official documentation for current function support and requirements.

Study checklist

■ Know how to extract year, month, day, quarter, weekday, and week number.

■ Practice adding months and finding month-end dates.

■ Understand the difference between date-part extraction and date-difference calculations.

■ Learn Excel working-day calculations and SQL Server DATEADD / DATEDIFF.

■ In DAX, understand date tables and the difference between date/time functions and time intelligence.